Model Description

Nomenclature

Indices	
i	Node
x	Generation technology
o	Conversion technology (PtX)
s	Storage technology
t	Period
Sets	
I	Nodes
X	Generation technologies
O	Conversion technologies
S	Storage technologies
T	Periods
Parameters	
C_x^{inv}	Annualised deployment cost of generation technology x
C_o^{inv}	Annualised deployment cost of conversion technology o
C_s^{inv}	Annualised deployment cost of storage technology s
C_x^{fixom}	Annualised fixed operation and maintenance cost of generation technology x
C_o^{fixom}	Annualised fixed operation and maintenance cost of conversion technology o
C_s^{fixom}	Annualised fixed operation and maintenance cost of storage technology s
C_x^{op}	Fuel and variable operation cost of generation technology x
C_o^{op}	Fuel and variable operation cost of conversion technology o
$\lambda_{i,t}$	Electricity price on node i in period t
$\overline{G}_{i,x}$	Maximum deployment of generation technology x in node i
$\overline{Y}_{i,o}$	Maximum deployment of conversion technology o in node i

$\overline{E}_{i,s}$	Maximum deployment of storage technology s in node i
$\underline{G}_{i,t}$	Minimum deployment of generation technology x in node i
$\underline{Y}_{i,o}$	Minimum deployment of conversion technology o in node i
$\underline{E}_{i,s}$	Minimum deployment of storage technology s in node i
$\rho_{i,x}$	Generation profile (solar or wind) in node i of technology x
$D_{i,t}^{act}$	Electricity active power demand in node i period t
$D_{i,t}^{reac}$	Electricity reactive power demand in node i period t
$D_{i,t}^{\mathrm{H2}}$	Hydrogen demand in node i for period t
$D_{i,t}^{\mathrm{Heat}}$	Heat demand in node i for period t
η_o	Efficiency and Coefficient of Performance of conversion technology o
$e_{i,s,t}^{loss}$	Storage losses of technology s, at node i in period t
ς_s	Storage losses factor of technology s
V_{nom}	Electricity system nominal voltage
$G_{i,j}$	Conductance of power line between node i and node j
$B_{i,j}$	Susceptance of power line between node i and node j
$\bar{S}_{i,j}$	Apparent capacity of power line between node i and node j
κ	Constant to constrain active and reactive power flow limits
τ^p, τ^q	Linearising parameters for active and reactive power components
α, β	Maximum and minimum ratios of reactive power generation to active power generation
PF	Demand power factor
π	Voltage angle limit
Variables	
$\chi_{i,t}$	Import of electricity from national grid at node i, for period t
$G_{i,x}$	Deployment of generation technology x in node i
$Y_{i,o}$	Deployment of conversion technology o in node i

$E_{i,s}$	Deployment of storage technology s in node i
$g_{i,x,t}^{act}$	Active dispatch of generation technology x in node i for period t
$g_{i,x,t}^{reac}$	Reactive dispatch of generation technology x at node i for period t
$g_{i,x,t}^{Heat}$	Dispatch of heat-only generation technology x in node i for period t
$y_{i,o,t}$	Dispatch of conversion technology o at node i for period t
$e_{i,s,t}^{ch}$	Charging of storage at node i, using technology s, for period t
$e_{i,s,t}^{dch}$	Discharging of storage at node i, using technology s, for period t
$e_{i,s,t}^{lv}$	Storage level at node i, using storage technology s, for period t
$\varrho_{i,t}$	Electricity curtailment at node i in period t
$p_{i,j,t}^{L}$	Electricity active power flow from node i to node j in period t
$q_{i,j,t}^{L}$	Electricity reactive power flow from node i to node j in period t
$V_{i,t}$	Voltage magnitude of node i in period t
$\theta_{i,t}$	Voltage angle of node i in period t
$\psi_{i,t}$	Reactive power compensation at node i in period t

Model Formulation

The objective of the model to minimise the system cost incurred due to the annualised costs of investments, operations, and electricity imports, as given in Eq. (1):

$$\min \sum_{\mathrm{i} \in \mathrm{I}} \Bigg[\sum_{\mathrm{x} \in \mathrm{X}} (C_x^{inv} + C_x^{fixom}) \cdot G_{i,x} + \sum_{\mathrm{o} \in \mathrm{O}} (C_o^{inv} + C_o^{fixom}) \cdot Y_{i,o} + \sum_{\mathrm{s} \in \mathrm{S}} (C_s^{inv} + C_s^{fixom}) \cdot E_{i,s} + \sum_{\mathrm{x} \in \mathrm{X}, \mathrm{t} \in \mathrm{T}} C_x^{op} \cdot g_{i,x,t} + \sum_{\mathrm{o} \in O, \mathrm{t} \in \mathrm{T}} C_o^{op} \cdot y_{i,o,t} + \sum_{\mathrm{t} \in \mathrm{T}} \lambda_{i,t} \cdot \chi_{i,t} \Bigg] \tag{1}$$

The first term describes the total annualised cost of investments, which is dependent upon the investment cost (C^{inv}) and fixed operational and maintenance costs (C^{fixom}) of the: generation (G), conversion (O), and storage (E) technologies capacities. Generation technologies G comprise electricity technologies and heat-only boilers, whereas conversion technologies Y include large-scale heat pumps for heat production and electrolysers for hydrogen production. The second term describes the operational cost of the different technologies that are associated with fuel and variable operational costs (C^{op}). These cost components are multiplied by the dispatch of

generation (g) and conversion (y). The final term provides the cost for electricity imports from the national grid, which is determined by the electricity price (λ) and the amount of electricity imported (χ).

The generation and storage capacities are determined by certain lower and upper limits of the different technologies, as described below.

$$\underline{G}_{i,x} \leq G_{i,x} \leq \overline{G}_{i,x} \tag{2}$$

$$\underline{Y}_{i,o} \leq Y_{i,o} \leq \overline{Y}_{i,o} \tag{3}$$

$$\underline{E}_{i,s} \leq E_{i,s} \leq \overline{E}_{i,s} \tag{4}$$

Equations (*2*)–(*4*) describe how the model allows for the use of a brownfield approach, taking existing installed capacity as the lower limit for the model to work with (indicated by underlining). In addition, the upper limit of the generation capacity (denoted with overlining) specifies the possibility of additional generation capacity. In the example of wind power generation, the upper limit refers to the maximum allowed capacity of wind power in a specific node i, which can be attributed to wind speed and land availability. Consequently, the operational dispatch of generation technologies is given by Eq. (*5*):

$$0 \leq g_{i,x,t}^{act} \leq \rho_{i,x} \cdot G_{i,x} \tag{5}$$

where the dispatch is limited by the installed capacity of generation technology while taking the specific generation profile into consideration. Examples of generation profiles are the wind speed in the case of wind power and the solar irradiance in the case of solar PV, whereas for thermal generation, the typical value would be 1, unless specifically mentioned otherwise.

Equation (6) describes that at every hour in every node, there must be a balance between the demand and supply of electricity:

$$\begin{gathered} D_{i,t}^{act} + \sum_{\mathrm{o} \in \mathrm{O}} \frac{y_{i,o,t}}{\eta_o} + \sum_{\mathrm{s} \in \mathrm{S}} e_{i,s,t}^{ch} + \sum_{\substack{i \in \mathrm{I} \\ i \neq j}} p_{i,j,t}^{L} + \varrho_{i,t} = \\ \sum_{\mathrm{x} \in \mathrm{X}} g_{i,x,t}^{act} + \sum_{\mathrm{s} \in \mathrm{S},} e_{i,s,t}^{dch} + \sum_{\substack{j \in \mathrm{I} \\ j \neq i}} p_{j,i,t}^{L} + \chi_{i,t} \end{gathered} \tag{6}$$

The left-hand side of the expression provides the demand part of the equation, which consists of the electricity demand, demand due to conversion to heat (power-to-heat) or hydrogen (electrolyser) that depends on the efficiency or coefficient of performance of the respective technologies, charging of electricity storage, electricity flow to other nodes, and curtailment if necessary. The supply part of the equation contains the generation dispatch from electricity generation technologies, discharge from electricity storage, electricity flow to node i, and the possibility of importing/exporting electricity. $\chi_{i,t}$ yields a positive value when it is importing and a negative value when it is exporting.

The nodal balances for the heat and hydrogen sectors are given in Eq. (*7*) and Eq. (*8*), respectively. Observe that there is no trade between nodes for both the heat and hydrogen sectors, i.e., the demand is met locally in the node.

$$D_{i,t}^{\mathrm{Heat}} + \sum_{\mathrm{s} \in \mathrm{S}} e_{i,s,t}^{ch} = \sum_{g \in \mathrm{G}} g_{i,x,t}^{Heat} + \sum_{\mathrm{o} \in \mathrm{O}} y_{i,o,t} + \sum_{\mathrm{s} \in \mathrm{S},} e_{i,s,t}^{dch} \tag{7}$$

$$D_{i,t}^{\mathrm{H2}} + \sum_{s \in S} e_{i,s,t}^{ch} = \sum_{o \in O} y_{i,o,t} + \sum_{s \in S,} e_{i,s,t}^{dch} \tag{8}$$

The expression that governs the storage balance is given in Eq. (*9*), where it ensures that the storage level at the next period is dependent upon the storage level, level of charging to the storage, level of discharging from the storage, and losses during the current period.

$$e_{i,s,t+1}^{lv} \leq e_{i,s,t}^{lv} + e_{i,s,t}^{ch} - e_{i,s,t}^{dch} - e_{i,s,t}^{loss} \tag{9}$$

where the storage losses is dependent on the storage technology losses factor ς_s and the storage level at the previous period, with storage losses factor obtained from [1]. Thus, storage losses is defined as

$$e_{i,s,t}^{loss} = \varsigma_s \cdot e_{i,s,t-1}^{lv} \tag{10}$$

As the model considers the electricity network infrastructure, each node is able to send or receive electricity that is accounted for in Eq. (6), given that the node has a power line connected to it. As the regional electricity infrastructure comprises AC power, it consists of active and reactive power components, where the relationship is non-linear. In other energy system modelling studies that have modelled the electricity flow [2,3], a simplistic representation of the electricity network, such as a copper plate model or a simplified DC load flow, has been used, which neglects the reactive power component. However, in the distribution grid, the electricity system is more-resistive compared to a transmission system, which means that the voltage could be more-affected by the electricity network. Thus, reactive power can no longer be neglected, as the voltage limit can be a limiting constraint in the model.

In the RESYST model, a linearised AC power flow approach derived from [4] is used, where this approach is intermediate between a full AC power flow, which captures the non-linearity of the power flow, and a DC power flow, which only considers the reactance of the network. A linearised AC power flow considers both active and reactive power components and applies the following assumptions to the electricity network: (1) the voltage angle differences between nodes are small; and (2) the magnitude of the voltage at each node is within the nominal value.

$$p_{i,j,t}^{L} \approx V_{nom}\left(-G_{i,j}\left(|V_{i,t}| - |V_{j,t}|\right) + V_{nom}B_{i,j}\left(\theta_{i,t} - \theta_{j,t}\right)\right) \tag{11}$$

$$q_{i,j,t}^{L} \approx V_{nom}\left(B_{i,j}\left(|V_{i,t}| - |V_{j,t}|\right) + V_{nom}G_{i,j}\left(\theta_{i,t} - \theta_{j,t}\right)\right) \tag{12}$$

Equations (11) and (12) approximate the active and reactive power flows, respectively. These equations govern the power flow from node *i* to node *j* at time *t*, accounting for the nominal voltage magnitude, network admittance, nodal voltage magnitude, and nodal voltage angle.

The active and reactive power flows between nodes *i* and *j* are limited to the apparent power-line capacity that connects nodes *i* and *j*, and it can be expressed as follows:

$$\left(p_{i,j,t}^{L}\right)^2 + \left(q_{i,j,t}^{L}\right)^2 \leq \left(\bar{S}_{i,j}\right)^2 \tag{13}$$

Nevertheless, one might notice that Eq. (*13*) is non-linear, as it describes a quadratic relationship between the active, reactive, and apparent power components in the AC power flow formulation. Thus, linearisation of Eq. (*13*) can be achieved through the following sets of equations, as previously described in [4]:

$$-\kappa \cdot \bar{S}_{i,j} \leq p^{L}_{i,j,t} \leq \kappa \cdot \bar{S}_{i,j} \tag{14}$$

$$-\kappa \cdot \bar{S}_{i,j} \leq q^{L}_{i,j,t} \leq \kappa \cdot \bar{S}_{i,j} \tag{15}$$

$$\tau^{p} \cdot p^{L}_{i,j,t} + \tau^{q} \cdot q^{L}_{i,j,t} \leq \kappa \cdot \sqrt{2}\bar{S}_{i,j}, \text{where } \tau^{p}, \tau^{q} \in \{1, -1\} \tag{16}$$

Observe that in Eqs. (14) and (15), the values of active and reactive power can either be positive or negative. This indicates that when the value is positive, the power flow direction is from node *i* to node *j*, whereas when the value is negative, the flow direction is the opposite. Equation (16) utilises the τ parameter, which limits the maximum active or reactive power delivery, according to its apparent power limit. Together, Eqs. (14)–(16) describe an operating region for both the active and reactive power forms that respects the non-linearity of the apparent power limits, as shown in Fig. 1. Nonetheless, this linearisation still allows areas for active and reactive power operating points that surpass the non-linear operating regions, displayed as areas of brown shading in Fig. 1, which shows the limitation of this approach.

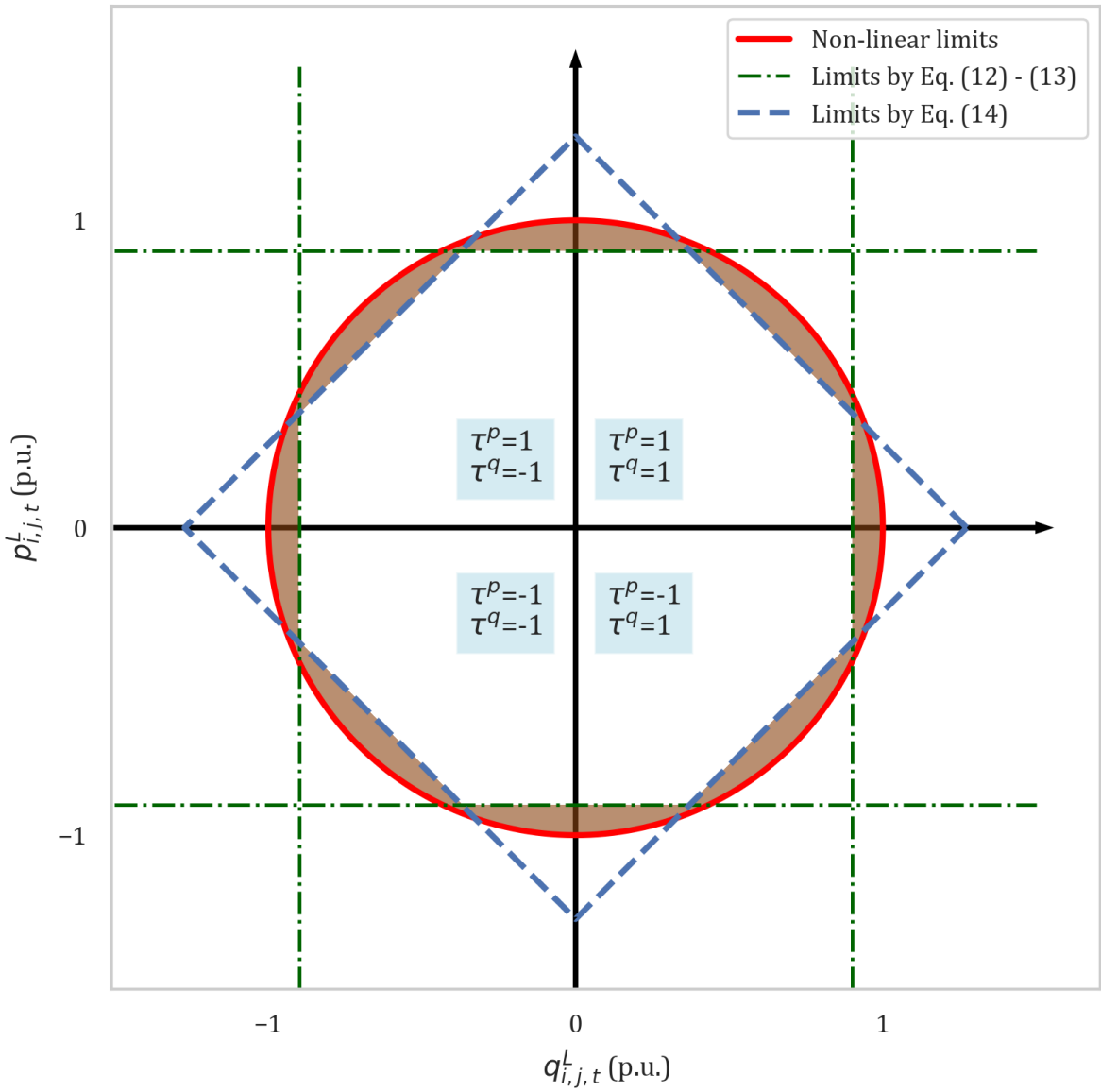

Fig. 1 Active, reactive, and apparent power limits.

As the model captures the reactive power, it is necessary to define a constraint that depicts the power balance for reactive power in each node at every time, as expressed by:

$$D^{reac}_{i,t} + \sum_{\substack{j \in I \\ j \neq i}} q^L_{i,j,t} = \sum_{x \in X} g^{reac}_{i,x,t} + \sum_{\substack{j \in I \\ j \neq i}} q^L_{j,i,t} + \psi_{i,t} \tag{17}$$

Compared to its active counterpart, Eq. (17) has fewer variables in the equation, where the reactive nodal balance consists of the reactive power demand and the reactive power delivery to other nodes in the demand side, while it takes into consideration reactive power generation by electricity generation technologies, reactive power delivery from other nodes, and nodal reactive power compensation in the supply side. Notice that since electric storage technologies cannot be used to supply reactive power, a reactive power compensation is used instead. The amount of reactive power demand can be computed according to the load power factor ($\cos\varphi$), where the load power factor is assumed to be constant for the entire model period:

$$D_{i,t}^{reac} = \frac{\sqrt{1 - PF^2}}{PF} \cdot D_{i,t}^{act} \tag{18}$$

The reactive power generation is limited by the amount of active power generation by a specific technology at a specific time:

$$-\alpha \cdot g_{i,x,t}^{act} \leq g_{i,x,t}^{reac} \leq \beta \cdot g_{i,x,t}^{act} \tag{19}$$

The values of α and β are dependent upon the electricity generation technology, which is classified according to whether or not a converter is needed to operate the technology. These values are taken from Swedish TSO Requirements of Generator [5], where it is stated that for converter-based technologies, the values of α and β are 1/3 and 1/3 respectively, whereas for non-converter-based technologies, the corresponding values are 1/6 and 1/3. Moreover, the model has limitations regarding the operating nodal voltage magnitude and angle, as given by:

$$\underline{V}_{nom} \leq |V_{i,t}| \leq \overline{V}_{nom} \tag{20}$$

$$-\pi \leq \theta_{i,t} \leq \pi \tag{21}$$

Then, the voltage angle difference between nodes is further limited by:

$$\theta_{i,t} - \theta_{j,t} \leq 0.5 \tag{22}$$

$$\theta_{i,t} - \theta_{j,t} \geq -0.5 \tag{23}$$

Eq. (20) enforces a limit as to how far the voltage magnitude can deviate from the nominal value; here, the limit is set at $\pm 5\%$ of the nominal voltage value. Equations (21)–(23) ensure that the voltage angle differences between nodes are sufficiently small, where in the model, the allowed differences are within 0.5 radians ($\approx 30°$). These sets of equations related to voltage are required so that the model abides with the assumptions mentioned previously, which allows the linearised AC power flow approach to be valid.

References

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